

Advanced (Habanero)

Q1. Solve the inequality $2x + 5 > 11$

- (A) $x > 2$
- (B) $x > 3$
- (C) $x < 2$
- (D) $x < 3$
- (E) $x = 3$

Q2. Graph the solution to $x - 2 < 5$ on a number line

- (A) $(-\infty, 7)$
- (B) $(-\infty, 3)$
- (C) $[3, \infty)$
- (D) $(-7, \infty)$
- (E) $(-3, 7)$

Q3. Write the solution to $x > 2$ and $x < 5$ in interval notation

- (A) $[2, 5)$
- (B) $(2, 5)$
- (C) $(-\infty, 2) \cup (5, \infty)$
- (D) $(-2, 5)$
- (E) $[5, 2)$

Q4. Solve the inequality $-3x + 2 > 5$

- (A) $x < -1$
- (B) $x > -1$
- (C) $x < 1$
- (D) $x > 1$
- (E) $x = -1$

Q5. Graph the solution to $x + 1 > 3$ on a number line

- (A) $(-\infty, 2)$
- (B) $[2, \infty)$
- (C) $(-\infty, 2)$
- (D) $[3, \infty)$
- (E) $(-2, \infty)$

Q6. Write the solution to $x < -2$ or $x > 2$ in interval notation

- (A) $(-\infty, -2) \cup (2, \infty)$
- (B) $(-2, 2)$
- (C) $[2, -2]$
- (D) $(-2, 2)$
- (E) $(-\infty, 2] \cup [2, \infty)$

Q7. Solve the inequality $x - 4 < 2x + 1$

- (A) $x > -5$
- (B) $x < -5$
- (C) $x > 5$
- (D) $x < 5$
- (E) $x = -5$

Q8. Graph the solution to $-2x + 3 > 1$ on a number line

- (A) $(-\infty, 1)$
- (B) $[1, \infty)$
- (C) $(-\infty, -1)$
- (D) $(-1, \infty)$
- (E) $(-\infty, 0)$

Open Ended Questions (FRQ)

Q9. Solve the inequality $x^2 + 2x - 6 > 0$ and graph the solution on a number line

Q10. Write the solution to $x + 2 > 5$ and $x - 1 < 3$ in interval notation and graph the solution on a number line

Chef's Tips

- Emphasize the importance of checking the endpoints of the intervals
- Encourage students to use a number line to visualize the solutions
- Remind students to consider the direction of the inequality when graphing the solution